Cambridge Assessment International Education

Cambridge O Level

MATHEMATICS (SYLLABUS D) **4024/11**

Paper 1 **May/June 2023**

MARK SCHEME

Maximum Mark: 80

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2023 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

© UCLES 2023

[Turn over

Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

© UCLES 2023

Mathematics Specific Marking Principles	
1	Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.
2	Unless specified in the question, answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.
3	Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.
4	Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).
5	Where a candidate has misread a number in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 mark for the misread.
6	Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

Abbreviations

cao	correct answer only
dep	dependent
FT	follow through after error
isw	ignore subsequent working
oe	or equivalent
SC	Special Case
nfww	not from wrong working
soi	seen or implied

© UCLES 2023

Question	Answer	Marks	Partial Marks
1(a)	154.3	**1**	
1(b)	24	**1**	
2(a)	$\frac{1}{3}$ oe	**1**	
2(b)	[0].25 oe	**1**	
3(a)	110	**1**	
3(b)	120	**1**	
4(a)(i)	$t - 5$	**1**	
4(a)(ii)	$2t$	**1**	
4(b)	19	**1**	
5(a)	$(3 + 5) \times 2 - 7$	**1**	
5(b)	$3 \times 5 - 2 + 7$	**1**	
6(a)		**1**	
6(b)	Sketch of regular hexagon with sides labelled 5cm	**1**	
7(a)	−21, −18, −10, −2, 17	**1**	
7(b)	−23	**1**	
8(a)	300	**2**	**M1** for $\frac{180}{3}[\times k]$ where $k = 1$, 4 or 5
8(b)	7.2	**2**	**M1** for *their* $\frac{a}{5} \times 12$ oe or $\frac{180}{3} \times 12$ oe or $\frac{\textit{their}\text{ total length}}{100}$
9	Rotation 90º clockwise oe [centre] (0,0) oe	**3**	**B1** for each
10(a)	$1\frac{5}{27}$ cao	**2**	**M1** for $\frac{4}{3} \times \frac{8}{9}$ or better

Question	Answer	Marks	Partial Marks
10(b)	$\frac{1}{12}$ oe fraction	**2**	**B1** for $\frac{11}{12}$ seen Or **M1** for $1-\left(\frac{1}{4}+\frac{2}{3}\right)$ oe
11	$x < -6$ final answer	**2**	**M1** for $-5-7 > 3x - x$ oe
12(a)	$\frac{7}{20}$ oe	**1**	
12(b)	70	**2**	**M1** for $\frac{14}{n} = 0.2$ oe
13(a)	336	**1**	
13(b)	1 : 200 000	**2**	**M1** for 4.5 : 9 × 100 × 1000 oe or **B1** for 1 cm : 2 km
14(a)	$9x - 1$ final answer	**2**	**B1** for answer $9x + k$ or $kx - 1$ Or **M1** for $15x - 10 - 6x + 9$ or better seen
14(b)	$2x^2 - 11x - 21$ final answer	**2**	**M1** for 3 of $2x^2$, $+ 3x$, $- 14x$, $- 21$
15	$6n - 5$ oe final answer	**2**	**B1** for $6n + k$ oe seen, any k
16	27	**3**	**M2** for $\frac{30 \times 360}{20^2}$ oe or **M1** for $\frac{x}{360} \times \pi \times 20^2$ [= 30 × π] oe
17	5.651×10^{26}	**2**	**M1** for $570 \times 10^{24} - 4.9 \times 10^{24}$ or $5.7 \times 10^{26} - 0.049 \times 10^{26}$ or **B1** for figs 5651 seen
18	[x =] $3y^2 - 2$ final answer	**3**	**M1** for squaring both sides **M1 FT** for correctly eliminating fraction following *their* first step **M1 FT** for correctly isolating x term leading to final answer Maximum 2 marks if final answer incorrect

Question	Answer	Marks	Partial Marks
19	$OA = OB$ equal radii $AT = BT$ equal tangents $O\hat{A}T = O\hat{B}T$ tangent perp to rad [Congruent] SAS Alternatively OT is common $O\hat{A}T = O\hat{B}T$ tangent perp to rad $OA = OB$ equal radii or $AT = BT$ equal tangents [Congruent] RHS Alternatively OT is common $AT = BT$ equal tangents $OA = OB$ equal radii [Congruent] SSS	**3**	**B2** for two pairs of equal sides/angles with correct reasons or **B1** for one pair of equal sides/angles with correct reason or for two appropriate pairs with no or incorrect reasons
20	$[f^{-1}(x) =] \frac{x-10}{7}$ oe final answer	**2**	**B1** for $7x = y - 10$ or for $\frac{y}{7} = \frac{10}{7} + x$ or for $x = 10 + 7y$ or better
21	60	**3**	**M2** for $12 \times 9 - 6 \times 8$ oe or for $12 \times 9 \times \left(1 - \left(\frac{2}{3}\right)^2\right)$ oe or **M1** for $9 \times \frac{2}{3}$ or 6 oe or $\left(\frac{2}{3}\right)^2$ oe or $\left(\frac{3}{2}\right)^2$ oe
22(a)	$(7 + 2x)(y - 3)$ final answer	**2**	**B1** for correct partial factorisation seen
22(b)	$3(a + 2b)(a - 2b)$ final answer	**2**	**B1** for $3(a^2 - 4b^2)$ or $(3a + 6b)(a - 2b)$ or $(a + 2b)(3a - 6b)$
23(a)	35 500	**1**	
23(b)	15 749	**3**	**B1** for 36 499 or 20 750 seen **M1** for *their* 36 499 – *their* 20750
24	3 columns correctly drawn	**3**	**B1** for each column After 0 scored, **SC1** for frequency densities of 2, 4 and 1.5 soi
25	$a = 3$ $b = \frac{4}{3}$ oe	**2**	**B1** for each

© UCLES 2023

Question	Answer	Marks	Partial Marks
26(a)	(1, 5)	**1**	
26(b)	$2y + 3x = 13$ oe 3 term equation	**4**	**M1** for [grad AB] = $\frac{7-3}{4--2}$ oe **M1** for $\frac{-1}{\textit{their}\ \text{gradient}AB}$ **M1** for substitution of *their* (1,5) into $y = (\textit{their}\ m)x + c$ oe or $y - y_1 = (\textit{their}\ m)(x - x_1)$ If 0 scored, **SC1** for sketch showing l is the perpendicular bisector of AB.
27(a)	$2\mathbf{a} + 3\mathbf{c}$ final answer	**1**	
27(b)(i)	$3\mathbf{c} - \mathbf{a}$ final answer	**1**	
27(b)(ii)	$2\mathbf{c} - \frac{2}{3}\mathbf{a}$ or $\frac{6\mathbf{c} - 2\mathbf{a}}{3}$ final answer	**2**	**M1** for a correct vector route along the lines on the diagram or $-2\mathbf{a} + \frac{2}{3}\textit{their}(2\mathbf{a} + 3\mathbf{c})$
27(c)	$\overrightarrow{AM} = \frac{3}{2}\overrightarrow{AT}$ oe and AM is parallel to AT and both go through A so ATM is a straight line **OR** $\overrightarrow{TM} = \mathbf{c} - \frac{1}{3}\mathbf{a}$ and $\overrightarrow{AT} = 2\overrightarrow{TM}$ and AT is parallel to TM and both go through T so ATM is a straight line	**2**	dep on correct $\overrightarrow{AM}$ and $\overrightarrow{AT}$ **B1** for $\overrightarrow{AM} = \frac{3}{2}\overrightarrow{AT}$ oe or for $\overrightarrow{TM} = \mathbf{c} - \frac{1}{3}\mathbf{a}$ and $\overrightarrow{AT} = 2\overrightarrow{TM}$ oe